

TASK 7

PCB DESIGN USING eSIM

12 V to 5 V Regulated DC Power Supply using LM7805

PCB Design Report

Designed using eSim / KiCad-based PCB Design Environment

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1. Project Overview

This project presents the design of a simple regulated DC power supply PCB developed using the eSim/KiCad-based PCB design environment. The circuit accepts a nominal +12 V DC input and uses an LM7805 linear voltage regulator to provide a regulated +5 V output. A 1N4007 diode is included at the input, capacitors are used for filtering and regulator stability, and an LED with a 330 ohm series resistor provides a visual power indication.

2. Objective

The objective is to demonstrate the complete PCB design workflow required in Task 7: schematic capture, component footprint assignment, PCB generation, component placement, routing, design-rule verification, 3D visualization, and generation/verification of manufacturing files.

3. Circuit Description and Working

The +12 V input is applied through connector J1 and passes through diode D1 (1N4007). Capacitor C1 (0.33 μ F) is connected at the input side of the LM7805 regulator. U1 regulates the input to approximately +5 V. Capacitor C2 (0.1 μ F) is connected at the output side for stability and filtering. The regulated output is available at J2. LED D2 and resistor R1 (330 ohm) form the output indication branch, while J3 provides the ground connection.

4. Schematic Design

The schematic was created in the eSim schematic editor. Electrical connections were completed before proceeding to footprint assignment and PCB generation.

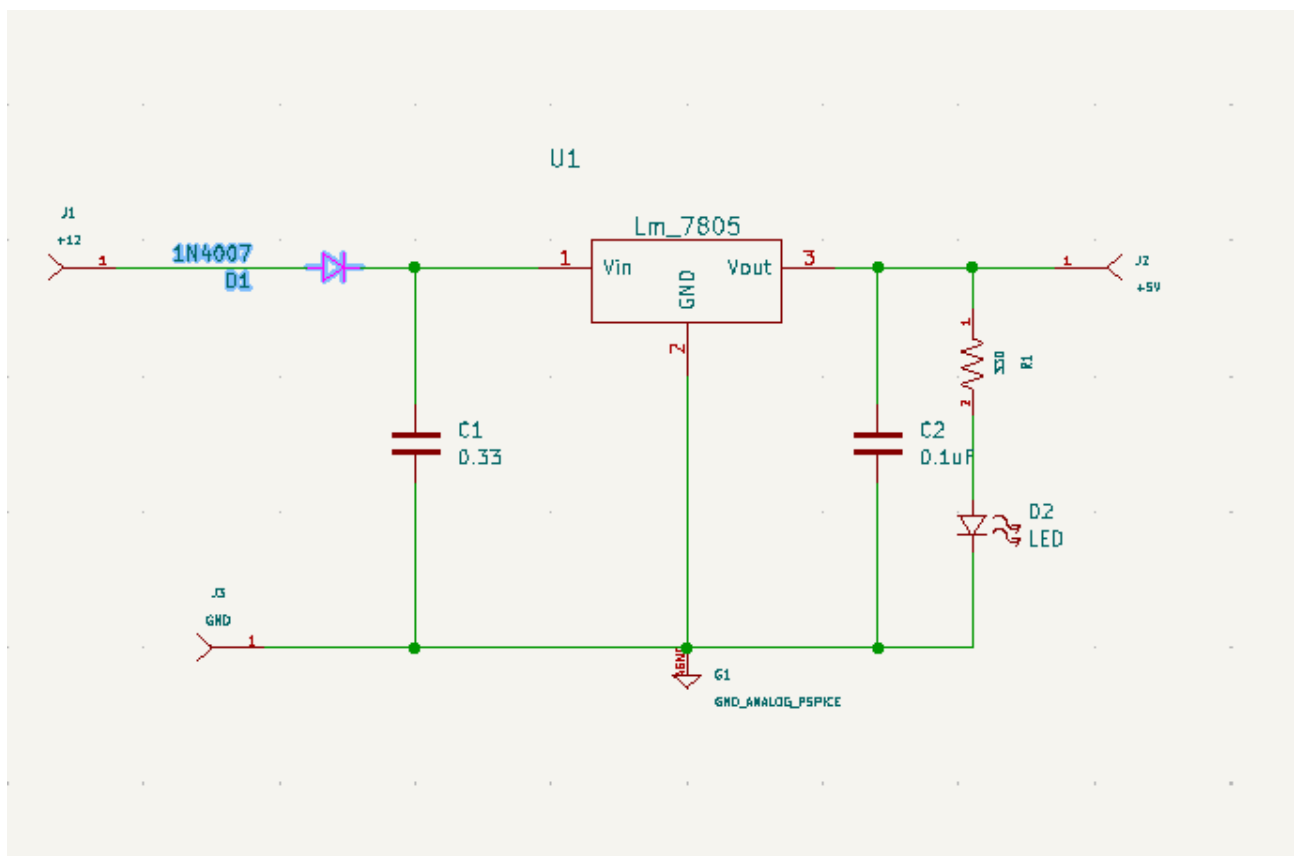


Figure 1: Schematic of the 12 V to 5 V LM7805 regulated power supply.

5. Footprint Assignment

Appropriate through-hole footprints were assigned according to the physical package and pin configuration of each component. The selected footprints are summarized below.

Reference	Component	Assigned footprint
C1	0.33 uF	Capacitor_THT:C Disc_D5.1mm_W3.2mm_P5.00mm
C2	0.1 uF	Capacitor_THT:C Disc_D5.1mm_W3.2mm_P5.00mm
D1	1N4007	Diode_THT:D DO-41_SOD81_P10.16mm_Horizontal
D2	LED	LED_THT:LED_D5.0mm
J1	+12 V	Connector_PinHeader_2.54mm:PinHeader_1x01_P2.54mm_Vertical
J2	+5 V	Connector_PinHeader_2.54mm:PinHeader_1x01_P2.54mm_Vertical
J3	GND	Connector_PinHeader_2.54mm:PinHeader_1x01_P2.54mm_Vertical
R1	330 ohm	Resistor_THT:R Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
U1	LM7805	Package_TO_SOT_THT:TO-220-3_Vertical

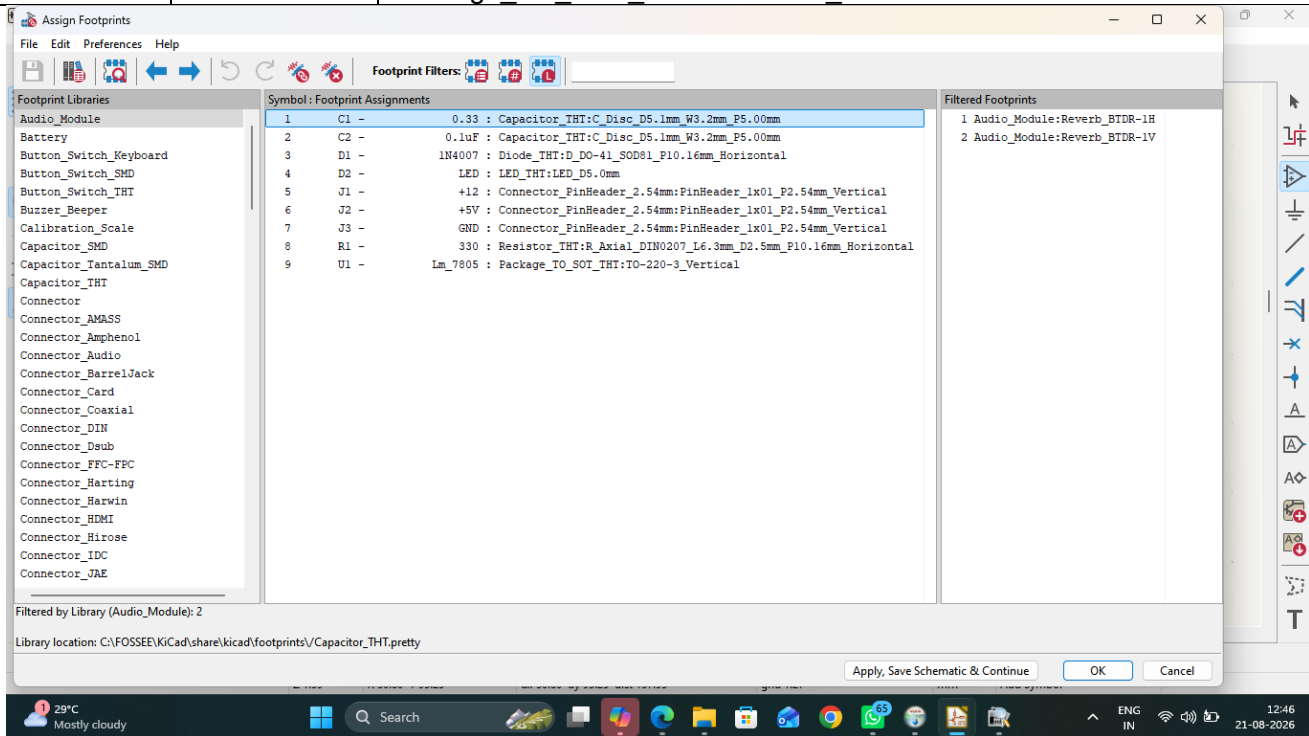


Figure 2: Footprint assignment window showing the selected component footprints.

6. PCB Layout and Component Placement

After updating the PCB from the schematic, the footprints were placed inside the board outline. The input components were arranged toward the left, the regulator and filtering components near the center, and the output/indicator components toward the right. This arrangement follows the signal and power flow and keeps the layout easy to inspect.

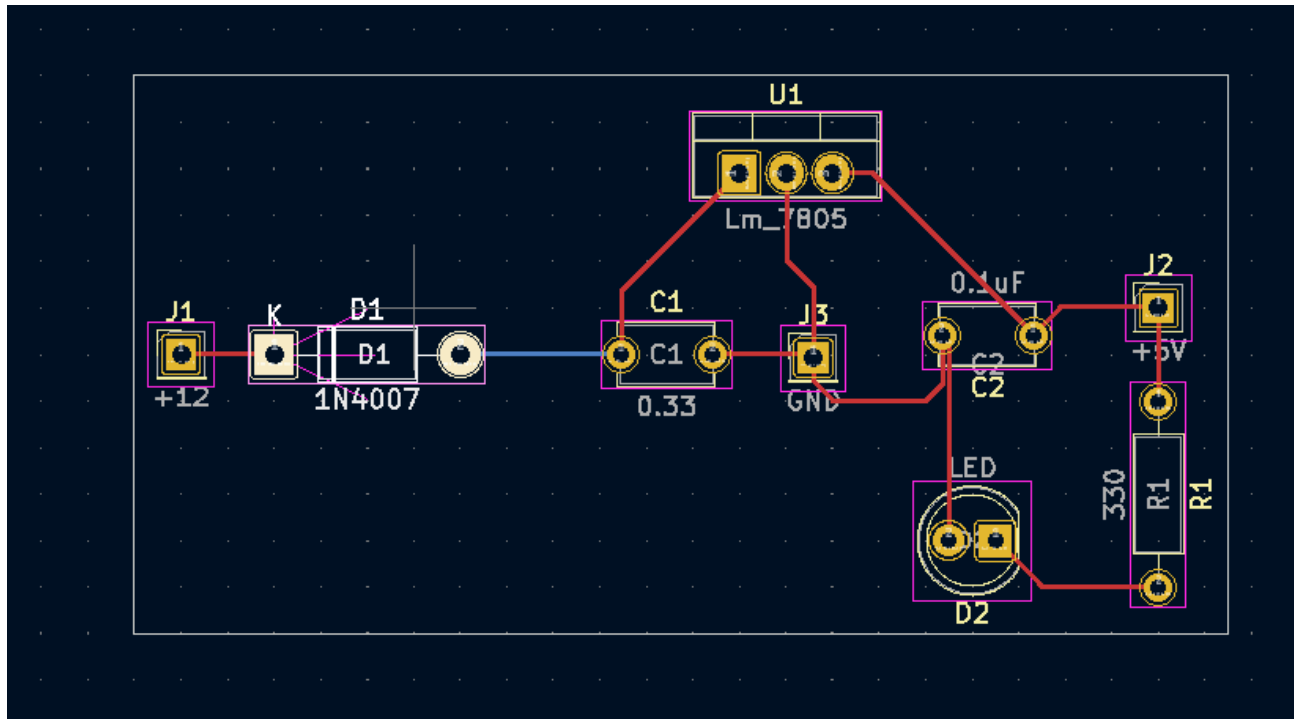


Figure 3: PCB component placement and routed connections.

7. Routing Methodology

Tracks were routed between the pads according to the schematic netlist. The routing was kept direct and organized, with the input, regulator, ground, output capacitor, output connector, and LED indicator connections completed without leaving unrouted nets. The board outline was created on the Edge.Cuts layer.

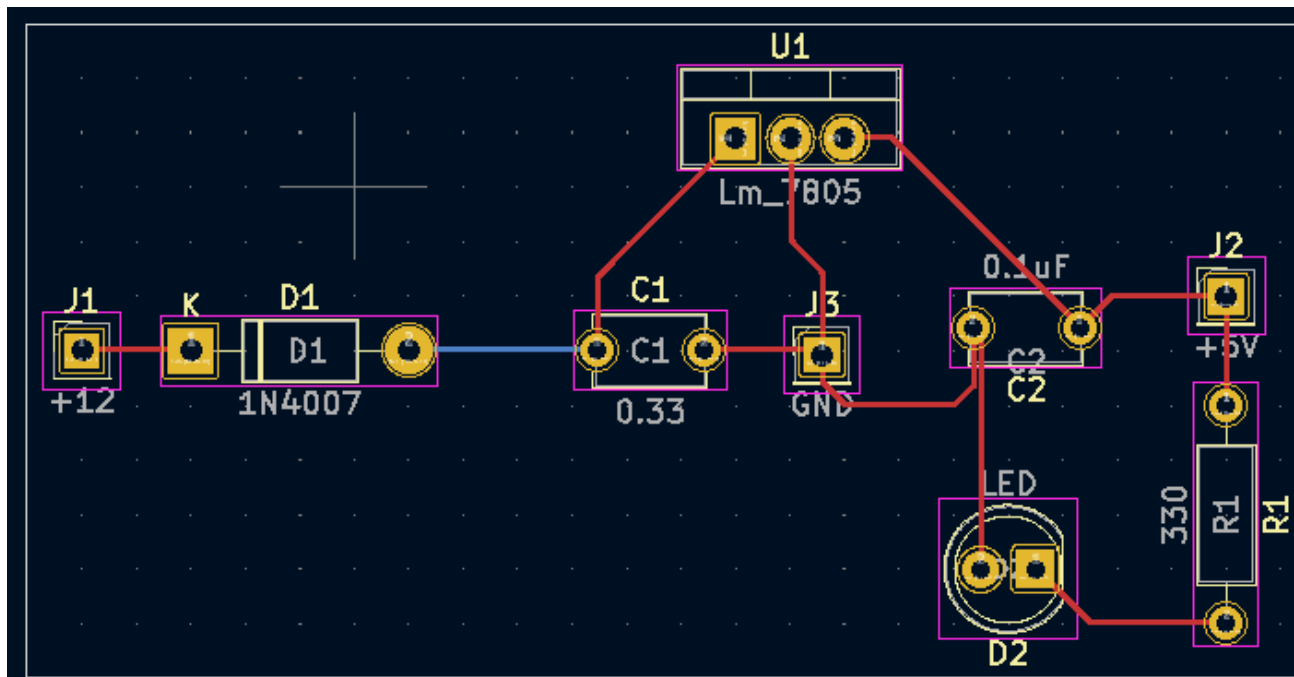


Figure 4: Final routed PCB layout.

8. Design Rule Check (DRC)

The PCB was verified using the Design Rule Checker. The final DRC result shows zero violations and zero unconnected items, with zero errors and zero warnings. This confirms that the routed PCB satisfies the configured design rules and that all required nets are connected.

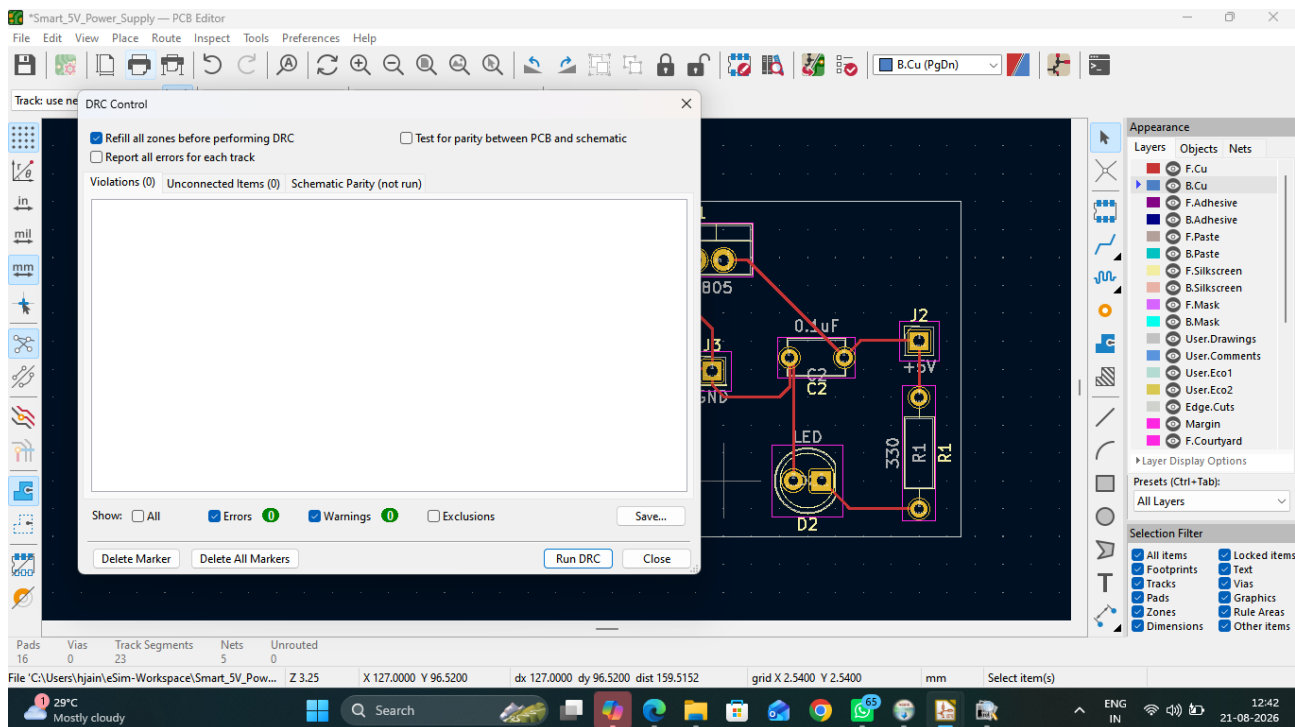


Figure 5: Final DRC result showing 0 violations, 0 unconnected items, 0 errors and 0 warnings.

9. 3D Visualization

The completed PCB was opened in the 3D Viewer to inspect the physical arrangement of the board and component footprints. The 3D view provides a visual check of component positions and the overall board form.

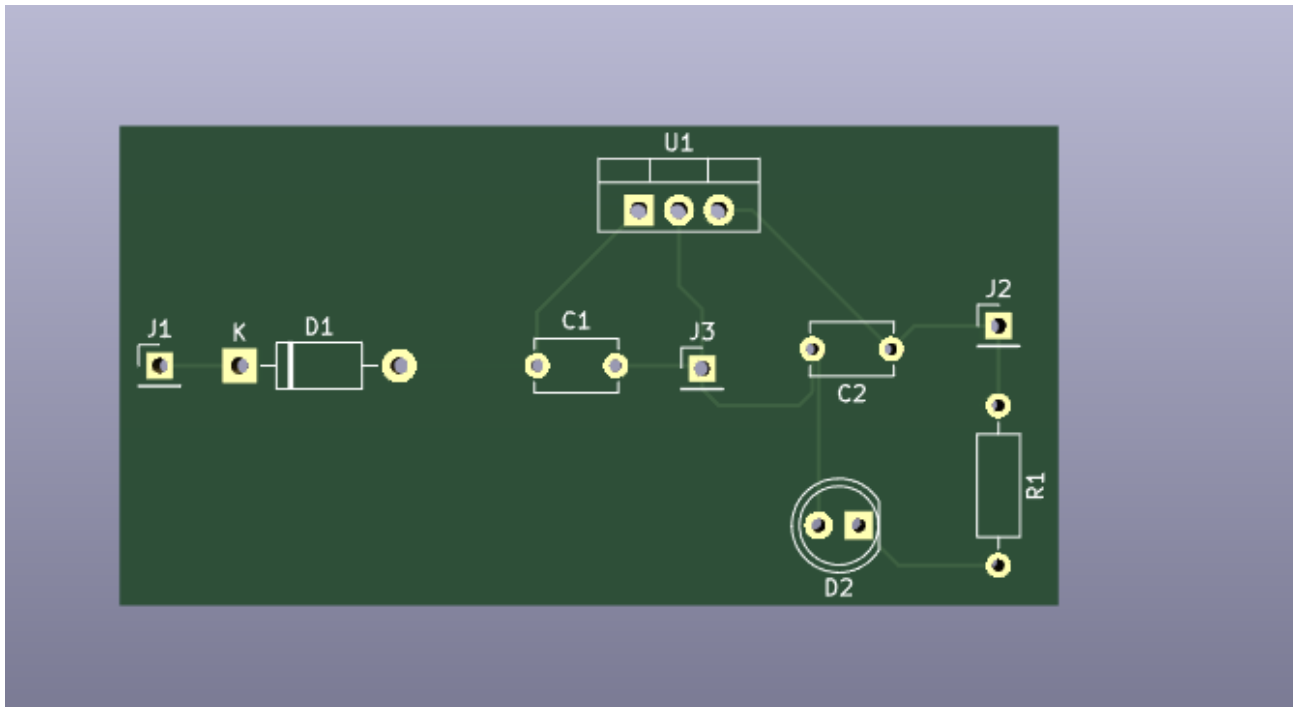


Figure 6: 3D visualization of the designed PCB.

10. Gerber and Drill File Verification

Gerber manufacturing files and drill files were generated after successful DRC. The generated layers were opened in the Gerber Viewer to verify the board outline, copper data, solder mask/silkscreen information, and drill locations before submission or fabrication.

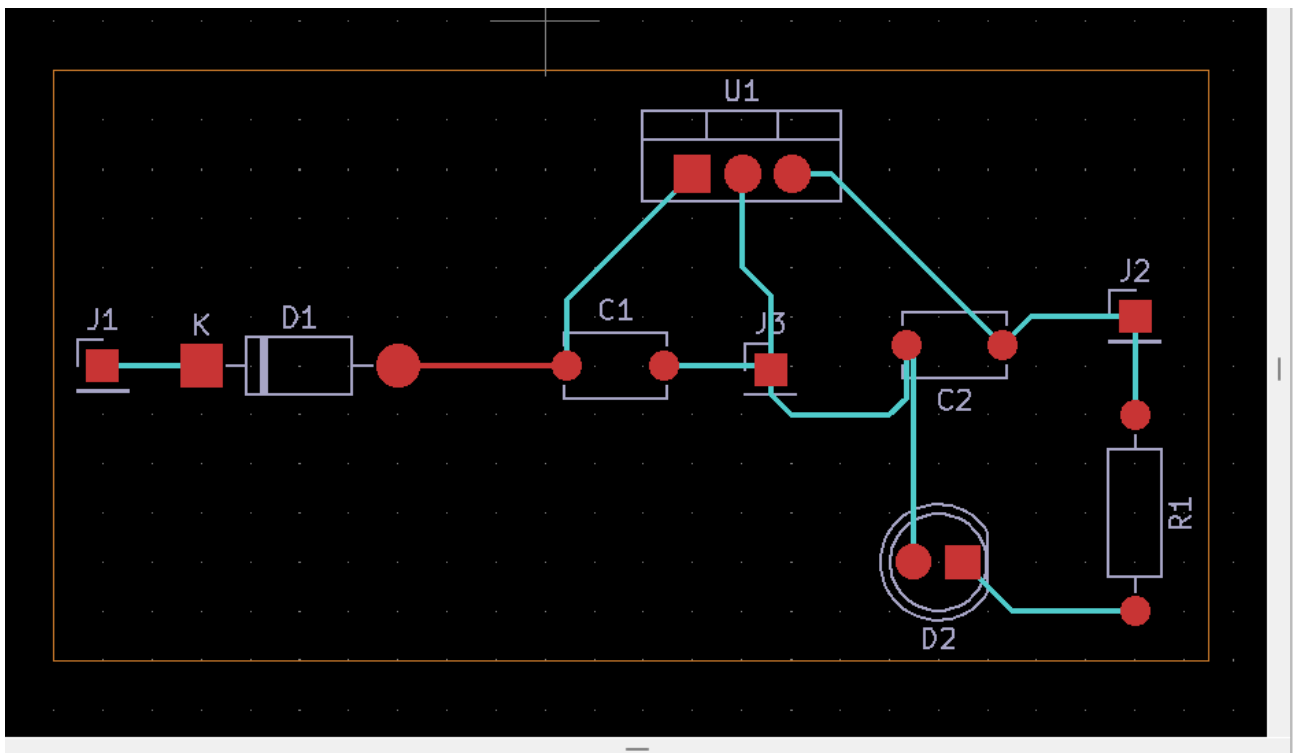


Figure 7: Gerber Viewer showing the generated PCB fabrication data.

11. Design Considerations

- The schematic was kept simple and organized to make the power path easy to understand.
- Through-hole footprints were selected for straightforward assembly and demonstration.
- Input and output capacitors were placed close to the LM7805 connections in the layout.
- The PCB outline was defined using the Edge.Cuts layer.
- Both front and back copper layers are included in the PCB/Gerber layer set as required for a two-layer board.
- DRC was completed before final manufacturing-file generation.

12. Result

A functional 12 V to 5 V regulated power-supply PCB design was completed through the required eSim/KiCad workflow. The schematic, footprints, PCB layout, routed connections, board outline, DRC verification, 3D visualization, Gerber files and drill files were produced successfully. The final DRC screenshot confirms zero violations and zero unconnected items.

13. Conclusion

The project demonstrates the complete PCB design flow from schematic capture to fabrication-output verification. The LM7805-based circuit provides a simple example for understanding footprint mapping, component placement, PCB routing, design-rule checking, 3D visualization and Gerber generation in eSim/KiCad. The design is ready to be packaged with the complete eSim project folder and supporting documentation for submission.

14. Files to Include in Final Submission

- Complete eSim project folder, including .kicad_sch, .kicad_pcb, .kicad_pro and any required project/library files.
- This detailed PDF report.
- Generated Gerber files and PTH/NPTH drill files.
- PCB design proposal.
- GitHub repository link and Pull Request link, as required by the Task 7 submission instructions.